→ Given a lattice of points R, a point G is a point in the reciprocal lattice iff $e^{i(G \cdot R)} = 1$

Reciprocal lattice is the set of all wave vectors G that yield plane waves with the periodicity of the given Bravais lattice

$R = n_1\bar{a_1} + n_2\bar{a_2} + n_3\bar{a_3}$

$G = m_1\bar{b_1} + m_2\bar{b_2} + m_3\bar{b_3}$

$a_i b_j = 2\pi \delta_{ij}$

$b_1 = \frac{2\pi(\bar{a_2} \times \bar{a_3})}{\bar{a_1} \cdot (\bar{a_2} \times \bar{a_3})}$

$b_2 = \frac{2\pi(\bar{a_3} \times \bar{a_1})}{\bar{a_2} \cdot (\bar{a_3} \times \bar{a_1})}$

$b_3 = \frac{2\pi(\bar{a_1} \times \bar{a_2})}{\bar{a_3} \cdot (\bar{a_1} \times \bar{a_2})}$

→ BCC: $a_1 = \frac{a}{2}(-\hat{x} + \hat{y} + \hat{z}) \ldots$

→ FCC: $a_1 = \frac{a}{2}(\hat{y} + \hat{z}) \ldots$

Reciprocal of FCC is BCC and vice versa

→ $S(r) = \sum_n \delta(r - a_n) \Rightarrow F(S(r)) = \frac{2\pi}{|a|} \sum_m \delta(k - 2\pi m/a)$

→ Lattice plane: plane containing atleast 3 non collinear points of a lattice

→ A plane (hkl) with intercepts x_i, y_i, z_i ; unit cell dimensions $a, b, c \Rightarrow h : k : l :: a/x_i : b/y_i : c/z_i$

$G = \frac{2\pi}{a}[hb_1 + kb_2 + lb_3]$ → reciprocal lattice vector normal to lattice plane with miller index [h k l]

→ two planes of indices [h k l] make intercepts on crystal axes of $na/h, nb/k, nc/l$ and $(n+1)\frac{a}{h}, (n+1)\frac{b}{k}, (n+1)\frac{c}{l}$

$\Rightarrow \frac{1}{d_{min}} = \left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 + \left(\frac{l}{c}\right)^2 \qquad d_{hkl} = \frac{2\pi}{|G_{hkl}|}$

→ X-Ray diffraction and Laue condn:

Constructive interference occurs iff change in wave vector b/w incident and scattered wave is equal to reciprocal lattice G $(k' - k = G)$

→ Brillouin zones:

1st brillouin zone = wigner seitz cell

N^{th} BZ = region in reciprocal space b/w n^{th} & $(n-1)^{th}$ Bragg planes

whenever Bragg plane touches one end of wave vector, diffraction condⁿ is satisfied, diffraction occurs whenever a wave vector touches a BZ boundary